

Design Doc Apr 2026

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! ! NOTE: This document only contains all the new systems or design updates since 4/1/2026

Other links:

Ruari's pitch on Verb Issue: [Poplife has an issue](#)

Building Upgrades

NOTES: I believe I heard about two different building upgrade last week, and I'm not sure which one are we implementing so I'm just gonna put both of them here.

Expand Building (by floor, fixed 2 buildings.)

There are 2 buildings map. Player by default has the F1 level, ground floor. Player unlock new area by floor, in sequence from close to further levels (Higher on ground, or lower in basement).

Expand Building (by room, customized placement)

There are multiple modular unit of rooms player can build freely on map in grid system similar to the shelf building system. Player need to connect them with floor connections, including escalators, elevators and portal, else the customers won't be able to access the area.

First Expand Mission (Tutorial)

First mission that guide player to expand their first area will be assigned when there's less than 3 spaces of floor left in the available areas. Before the first expand mission is assigned, player will not see (hide)/ access (locked) this part of the game mechanic.

Shelf Attributes

 **POPLIFE NARRAIVE SHEET** tab/ ITEMS IN GAME

*** Column B, C, D show whether they're in any bundle/ quest requirement/ quest rewards. It's automatically updated.

List of Attributes:

Attribute	Definition	Formula
Display Name	Name displayed and referenced in all related sheets. Used as Item codes in Unity Project too.	
Brand	The brand of the product. Brand affects the visual style of the shelf. Brand should be a part of the narrative of quests/ shelf descriptions.	
Category	The type of product. Some events and policies are based on it.	
Level	1 to 5, 1 as default. Can be upgraded by fame. Level up raises Sell price , Profit , and Appeal but also raises Stock cost and Shelf Maintenance .	

Building Cost	Money cost to build the shelf	
Upgrade Cost	Fame cost to upgrade shelf level	Floor (build cost * 0.5 + level ² *5)
Sell Price	Money customer pay the player to buy one product.	Stock fee + profit
Stock fee	Money player pay to stock one product.	Floor (Build cost *0.05 + current level ^ 2 * 0.5)
Profit	Money player will actually earn by selling one product	Floor (build cost * 0.03 + current level ³ *0.25)
Maintenance	Money cost by having the shelf in store everyday.	Floor(0.9 * profit + 3 * profit ^{0.5})
Appeal	Appeal of individual shelf	Floor(build cost * 0.09 + 2 * level ^{1.5} *0.2)

Profiting

The shelf now

Manage Stock

The player needs to adjust amount of stock for each product at the start of the day. Player adjust it with a scroll bar for each product.

Scroll bar contains:

1. Stock of player's choice
2. Recommended stock. (Sale of this product yesterday (D-1), Sale the day before yesterday (D-2), etc.)

a. Weighted average formula:
$$= 0.5 \times \text{昨天} + 0.3 \times \text{前天} + 0.2 \times \text{大前天}$$

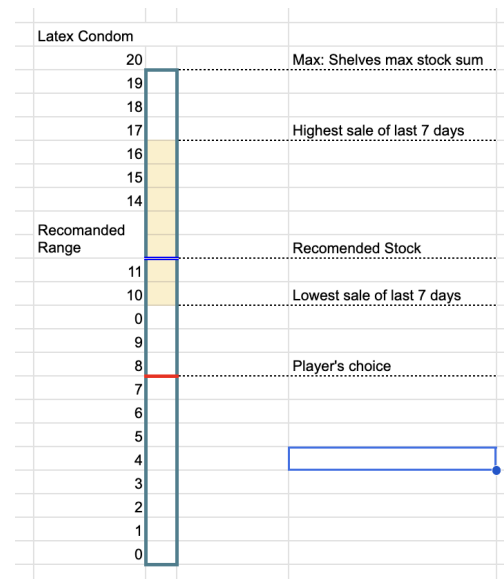
$$= 0.5 \times (D-1) + 0.3 \times (D-2) + 0.2 \times (D-3)$$

NOTE: This one makes no sense when the sale keeps growing or declining in a few consecutive days.

- b. Linear regression formula 线性拟合: PopLifeDesignDoc

NOTE: This one makes sense to me, but might has performance issues.

4. Range of recommended stock.



A range of stock recommendations. The highest and lowest are the highest and lowest in the past 7 days. This is a colored block in UI.

NOTE: Customer upgrade should be much easier, and their money bag should grow much more each lv.

Customer's decision logic should be weighted by both preferred shelves list AND shelf appeal. My idea: For each customer, their preferred shelf's appeal should be *3 for them, but they still buy product on appeal based decision making. We also want to make sure product's won't be sold out by sequence of appeal rank, so we might want to take *Shelf's Distance to customer* into account. We can simply make it absolute distance to player, with those in the same floor a boost. We can move the player when they first enter the store, by directing them to one of their fav shop first.

Upgrade

Move Shelf Update

Move shelf now cost fixed amount of money, 150 for now.

(Strategy: this feature would be more used in later stage of game, which focus on bundle making and shelf upgrading. In the early stage, where player go through tutorial, get familiar with build, quest, stock, lottery and other basic game mechanic, they don't need to worry about it, can't afford it anyways.)

Store Stock

There will be a limit of total stock that player can hold at a time. At the end of the day, products that're not sold will stay in the warehouse, taking up places for new products.

Upgrade: New Shelf

When new shelf is built, store stock limit will += lv.1 stock of this shelf. When it's removed, store stock will -= lv.1 stock of this shelf.

Strategy: When shelves are upgraded, player won't be able to fully stock all shelves everyday, so they have to stock shelves strategically or earn more store stock limit.

There are debuffs and buffs on categories, so player will want to stock different kind of shelves each day, creating meaningful action with decision making.

NOTE: IDEA!!! Add a "Trending Categories" and "Trending Brands" for each week. We can randomize it, but give player a lot of hints before the week it's activated through broadcast!!! (We can also add categories or brands that's disliked by the market for a while) reference: Kairosoft cakery game

Bundles

 POPLIFE NARRAIVE SHEET tab/ bundles

Bundles consists of multiple shelves. It will automatically activate when the shelves are placed together. Customer will buy all the items at once if they buy one of the bundle. This mechanic gives player a strategy to sell more low appealing products. Information of activated bundles information are available in Store Stats.

NOTE: we should design some sort of visual effect on the shelves when a bundle is activated. Could be some sparkling particles.

Recipe (Bundle Blueprint)

Each bundle is made of 2 - 4 shelves. Recipe of bundles can be unlocked as a prize from lottery.

Information of unlocked recipe of bundles are available in alanbot.

Effect

Each bundle has effect on each product, to lower stock price and price.

Shelves of different level in a bundle, will be automatically upgraded to the level of the highest level shelf in this bundle. This effect will be removed when the shelf is moved out of the bundle.

Alt Idea on bundle:

Player can create bundle. There are designed "matching" pair of products. If the bundle have them, it will have a higher boost in stock price, profit, appeal and maintenance fee. If not, the bundle only have a tiny amount of boost, about 1 - 2. (Referencing kairosoft cakery/ ramen/ ... games)

Quests

 POPLIFE NARRAIVE SHEET tab/ quests

Quest has an assigned condition, a complete condition, and rewards.

Assigned condition & Complete condition

Customer/ NPC:

Each quest is assigned by a customer or NPC. For those assigned by customers, the quest will only be assigned when this customer is unlocked.

Quest Type: Required shelf

Each quest has 1 required shelf. Player complete this shelf by build this shelf. Quest will only be assigned when this shelf's blueprint is unlocked.

This type of quest will be assigned when the customer and blueprint are both unlocked

Quest Type: Others

Other types of quests include many different complete conditions that works like achievements. This kind of quest is always activated, and completing them leads to a higher level of the same quest. The conditions include:

Day loop quest

1. earn xxx money/fame in a day/ week
2. Have xxx customer in a day
3. Have xxx purchases in a day

Life time quest

4. earn xxx money life time
5. place xxx shelves
6. Hold xxx event
7. Unlock xxx blueprint
8. Complete xxx customer question
9. Complete xxx customer quest

Reward

Rewards are usually Blueprint, new Customer, Customer exp, Bundle recipe, Money, and Fame. Multiple rewards for one quest are possible.

Customer Question

Shelf-based random question selection

Day Loop

Morning - Player Clicked on Open Store

Stock phase

Policy update

Store Hour - Store Runing

Store Operating

Store Hour - Store Closed (automatic)

End of the day panel

NPC Quests

Main Story Line

****End of week:**

Lottery

Store Policy (Replace Player held event)

Shelf policies are effects that can be turned on and off by the player at the start of each day (OR ANYTIME???). This effects on one aspect has cost on other aspect.

Unlocking Condition:

Policies can be unlocked in lottery.

NOTE: where else? What about some quest? Or some sort of policy shop like the blueprint shop?

List of Policy:

Policy	Effect	Cost	Strategy
Extended Store Hours	Extend the store hour of the game by 3 hours.	Increase total shelf maintenance fee by 10%	Useful at earlier stage, cuz maintenance fee is relatively cheap Or later stage, when player has a large customer base.
Category/*Brand* Sale	For all shelves in the category/ brand: Appeal * 1.5 Price * 0.8	//Lower profit	Sales policies are useful when the trending products are not trendy anymore, and player need to remove them from stock for new trending products.
Category/*Brand* Super Sale	For all shelves in the category/ brand: Appeal * 2.3 Price * 0.5	//Much lower profit/ no profit	
Advertisement	Total appeal * 1.3	Increase total shelf maintenance fee by 10%	**Maybe Good when Low shelf levels
Eco-friendly Business	Total appeal * 1.3	Increase all stock fees by 1	Bad at early stage, Useful at later

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Mechanics

1. Resource

- 1.1. **Game time:** In-game time that flows automatically. Game time pauses when player activates any UI Panel. Actions in the game consume time to finish, including construction, customer actions, etc.
 - 1.1.1. **Time Speed:** The Speed of game time can be controlled by player. Player can choose to pause, 1x or 2x the time flow.
- 1.2. **Money:** The main currency in the game. Player earns money from operating stores, completing requests, etc. The player uses money to construct.
 - 1.2.1. **Bankrupt:** At the start of Daily Settlement, if money < 0, the player will lose and need to restart from the latest checkpoint, usually the day before.
- 1.3. **Fame:**
A consumable resource, a second currency in game. Earned by completing requests (significantly), and in daily settlement (subtly, at the cashier), and in daily achievement. Used to upgrade product category level, buy blueprint, hold promotion event, etc. Harder to gain in the later stages.

2. xPlayer Goal

- 2.1. **Store Star Upgrade:** The store can be upgraded from 0 stars to 5 stars.
 - 2.1.1. **Requirement:** Each upgrade requires certain requirements, and the player can upgrade in an info panel when the requirements are all met. The requirements are usually quotas to meet, not resources to be cost. For example, total shelves built, total customer amount, total shelf level, total customer level, highest historical daily income, etc.
 - 2.1.2. **Reward:** Each success upgrade rewards the player with a large amount of *money* and *fame*, more *space*, and higher *shelf* and *customer capacity*.
- 2.2. **Main Mission** (VIP Story Mission): VIP Story Mission is the main storyline and main missions in the game. It's a customer request held by VIP customers, a kind of expanded customer service. The requirements of VIP Missions are not explicit, and players need to solve it as a puzzle through narrative context. There will only exist one active VIP Story Mission at the same time.
 - 2.2.1. **Story:** Each VIP Story Mission has an introductory story that serves as the context of the mission's requirements. The story is shown in the form of dialogue between VIP customer and player. (See 7.7)

- 2.3. **Quest:** Quest is given to the player at the end of each day, before daily settlement from the NPC.
 - 2.3.1. **Unlock Condition:**
 - 2.3.2. **Complete Condition:**
 - 2.3.3.
- 2.4. **Daily Settlement:** Daily settlement count and visualize the data changed during the day, including money and fame earned, served customers, completed and uncompleted achievements etc. (See 3.4.7.2)
 - 2.4.1. **Daily achievement:** Daily achievements are tiny, random goals for the player to accomplish. There will always exist three unfinished achievements, and if any of them are achieved, new achievement requirements will be assigned randomly from the database. The status of Daily achievement is displayed during the daily settlement. (See 3.4.7.2)
- 2.5. **Calendar events:** The calendar is an organized view of trends, events, and quest goals happening each day chronologically. This information will be announced first by the robot or in the newspaper in the morning.
 - 2.5.1. **Trend:** Trend is a buff effect on customers' preferences and need for specific merchandise, event, or category. Trends usually last for a day.
 - 2.5.1.1. Holiday
 - 2.5.2. **Event:** The event can be held by the player at the event hall. Event attracts customers, and has special buff effects similar to the trend.
 - 2.5.3. **Quest Goal:** Deadlines of quests received from npcs will also be pinned on the calendar as events.

3. Store

- 3.1. **Map:** Wide space for buildable modular areas in units of Areas. Areas are stacked and can be connected in an aesthetically irregular manner. The player can pay money to construct new areas. Beside new space for shelves, new areas also increase customer capacity,
 - 3.1.1. **Area:** Placement on the floor is grid-based. Each floor is 2-grid height. Each column can only place one shelf. Decoration and Floor connections (stairs and elevators) are placed, but don't occupy the grid.
 - 3.1.2. **Customer Capacity:** The limit of the number of customers that can exist in the store at a time.

- 3.1.3. **Unlock New Area:** With a lot of money.
- 3.2. **Shelves:** Placed on the floor, where customers buy products. Shelf attributes:
 - 3.2.1. **Price:** The customer pays the player this money per product.
 - 3.2.2. **Maintenance Fee:** Money that's reduced at daily settlement.
 - 3.2.3. **Stock:** The maximum number of products that can be bought in one day. If stock = 0, the shelf is sold out and shut down for the rest of the day. Players decide how much to restock every day.
 - 3.2.4. **Attractiveness:** The more attractive the shelf is, the more possible for a customer to come and buy its product. Shelf attractiveness together is the store's total attractiveness.
 - 3.2.5. **Category:** Customer will have an interest level in each category. The customer will visit the shelf from the highest interest level to the lowest. A category can be upgraded from Lv 1 to Lv 5 with a lot of fame and money, and higher levels will need an expensive maintenance fee. Category-level upgrade significantly boosts the attractiveness of all shelves in the category. Category includes: Lingere, Condom, Vibrator, Fleshlights, Lubricant
 - 3.2.6. **Build Fee:** A one-time money cost when placing the building.
- 3.3. **Facilities:** Boost floor attributes or have special use. (A short list with varied effects.)
 - 3.3.1. **Facilities attributes:**
 - 3.3.1.1. Effect
 - 3.3.1.2. Build Fee
 - 3.3.1.3. Maintenance Fee
 - 3.3.2. **Facilities examples:**
 - 3.3.2.1. **Cashier:** Each floor needs at least one cashier. Starting from the second unlocked floor, each cashier needs at least one employee.
 - 3.3.2.2. **Air Conditioner:** Reduce embarrassment Level for all customers on the same floor.
 - 3.3.2.3. **ATM:** Customer with a high interest, low embarrassment, and low money bag will use the ATM to restore their money bag.
 - 3.3.2.4. (...)//

- 3.4. **Blueprint:** Shelves and Facilities need blueprints to be built.
- 3.4.1. **Unlock:** Blueprints are unlocked when the player reach a specific day in game, or as reward of quest, or customer level.
- 3.5. **Construction Mode:** The player can place, destroy, and move shelves and facilities. Item placement is grid-based.
- 3.5.1. **Place:** Costs a build fee and consumes one blueprint. Construction time: 15 seconds.
- 3.5.2. **Move:** Cost movement fee (small amount). Construction time: 5 seconds
- 3.5.3. **Destroy:** Return one blueprint. No money cost or returned. Instant construction.
- 3.6. **Day Loop:** A Game time day costs 30 seconds in real time. A day counter UI in the corner counts the date in the format: Day xx.
- 3.6.1. **Store Hours:** 12:00 to 23:00. Time flows at normal speed. Customers AI will move and buy products in the store during store hours. The player can activate any UI. The store hours can be upgraded at a mid-stage of gameplay.
- 3.6.1.1. **Daily Settlement:** UI panel pop-up with information about the last store hours, at 10 pm. During daily settlement, the time flows quickly to 11 am, the next day, then pauses
- 3.6.1.1.1. **Profit - Money**
- Today's Sale
Maintenance Fee
Money Change
Lifetime Income
Lifetime Expense
- 3.6.1.1.2. **Profit - Level Up**
- 3.6.1.1.3. **Other Data**
- Fame Earned
Customer Entered Today
Daily achievement (See 2.5)
Competition / News / ...
- 3.7. **Trend:** Every day has a different Trend, affecting the attractiveness of products.
- 3.7.1. **Trend Calendar:** The trend calendar records each trend in the game. It's shown to the player at the start of the day, and the player can always check it during gameplay.

- 3.7.2. **Store info panel:** A UI panel that keeps track of all store information.
Store level: A store-level challenge from 0 stars to 5 stars. The player needs to fulfill specific requirements to upgrade the store level. Upgrading the store level will unlock blueprint and customer, reward money and fame, etc.
Total store count
Total attractiveness count
Total income count
- 3.8. **Stat:** A UI panel that holds all ongoing information in the day.
 - 3.8.1. Economy
 - 3.8.2. Shelf
 - 3.8.3. Current Customers
 - 3.8.4. Current Customer list: A list of customers in the store, updating dynamically.
Can be organized by loyalty level, or total expense; ascending or descending.
 - 3.8.4.1. List of customers with four column
Loyalty Level
Icon
Name
- 3.9. **Stock:** The maximum number of products that can be bought in one day.
 - 3.9.1. **Restocking:** Performed manually by the player (or automatically / semi-automatically during daily settlement). Player controls the restock of each product by a scroll bar. Previous restock amount

4. Customers

4.1. Customer Attributes

- 4.1.1. **4 Interested Shelves:** Each customer has 4 interested shelves. When the customer arrives, they will first buy the 4 interested shelves in a random sequence. After that, customer will head to the most attractive shelves.
- 4.1.2. **Money bag (MB):** When customers use up all the money, they will leave.
- 4.1.3. **EXP:** When EXP is full, upgrade the customer and empty EXP. EXP is gained from completing requests and customers making purchases.

- 4.2. Customer Upgrade:** Customers are automatically upgraded to the next level when the EXP reach condition. Upgraded customer have larger money bag.

4.2.1. Customer Levels

Customer Type	EXP cost	Money bag		
0	10	10		
1	30	30		
2	50	50		
3	100	90		
4	200	120		
5	500	180		
Max	1500	300		

4.2.2. Customer AI:

- 4.2.2.1. **Spawn:** Customer spawns randomly in a random interval of time, until amount of customers in store reach the capacity limit.
- 4.2.2.2. **Buy Amount:** The actual amount of product the customer will try to purchase at one shelf. Buy amount is one at default.
- 4.2.2.3. **Buy List:** A list of shelves that the customer will probably visit.
- 4.2.2.4. **Target:** The Customer will always move towards the target. Can be shelf, facility or exit.

4.2.3. Customer AI Logics

- 4.2.4. Customers randomly spawn at the entrance.
- 4.2.5. Set the target to the most attractive shelf for this customer.

The selection logic:

1. Randomly select 1 out of 4 interested shelves. Next, randomly select 1 out of the other 3 interested shelves. Repeat 4 times until all interested shelves are visited.
 2. Then, visit all shelves in the store from the highest attraction level to the lowest. Skip the shelves already bought (4 interested shelves)
 3. If money bag is not enough to buy it, go to cashier and make payment.
- 4.2.6. Move to the target shelf. Buy action. customer money - -
 - 4.2.7. Buy amount =
 - 4.2.8. Shelf stock -= buy amount.
 - 4.2.9. Make Payment: Customer go to cashier. Player Money ++
 - 4.2.10. Payment = (buy amount) * (shelf price). Customer's money bag -= payment. Player money += payment.
 - 4.2.11. Empty player inventory. Remove the bought shelf from the customer's buying list.
 - 4.2.12. Move to step 2, loop.
 - 4.2.13. *Customer AI Notes:*

4.2.14. Customer Actions:

Walk speed = [unfinished, need tests]. Customer speed is not changeable by the player.

Action	Time cost (seconds)	Notes
Move	Distance/speed	
Wait	(line position) * (checkout/ buy time)	
Buy	1 + (buy amount * 0.1) second	
Checkout	1 + (buy amount * 0.1) second	

5. Storytelling

1 Story Summary

[unfinished]

2 Background Setting

[unfinished]

3 Storytelling Tools

Story in the game will be narrated mainly through VIP Story Mission. Flavor text will help with world-building in Customer Request, Shelf info panel, Customer info panel, etc.

We are using Dialogue Tree in [Node Canvas](#) to implement dialogues into gameplay.

1 Tutorial Dialogue:

[Link to Tutorial Dialogue.](#)

[unfinished]

2 VIP Mission Dialogue:

[Link to VIP Mission Dialogue.](#)

[unfinished]

Art

6. Art [unfinished]

1 Artstyle/reference

2 Character

3 Store

4 Scene

5 Technical Art

...

7. UI [unfinished]

[Link to UI Design Page \[Figma\]](#)

7.1 General UI

7.2 Runtime Indicator UI:

Pop Bubble: Overhead, indicating customer action, including getting products from shelves, purchasing, thoughts, and emotions, etc.

Broadcast: A one-line message about each customer's Purchase, rolling horizontally. Example format of each message: [customer name] bought [amount] of [product], + [sales money].

Messages: A list of messages about important notices. Text will be in three colors: Gold, White, and Red.

- Gold: Notification of the occurrence of something exciting
 -
- White: Notification of the occurrence of something normal
- Red: Warning messages of the occurrence of something negative.
 - Customer leaves because no interesting shelves available
 - ...

Unlock/ Upgrade/ important, etc.

7.3 Statistic Panel:

Daily Settlement Panel:

7.4 Construction Panel:

7.5 Information Panel:

7.5.1 Customer

7.5.2 Shelf

7.5.3 ...

7.6 Request:

7.6.1 Request List

7.6.2 Request Detail

7.7 VIP Mission:

8. Audio [unfinished]

8.1 Music

8.2 SFX

8.2.1

8.3 Audio Implementation

...

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[Link to the core loop](#)

[Link to Spreadsheet](#)

[Link to Economy System Doc and Notes](#)

Concept Definitions

1 Resource

[Implemented]

1.1 Game time: In-game time that flows automatically. Game time pauses when player activates any UI Panel. Actions in the game consume time to finish, including construction, customer actions, etc.

[Not Implemented]

1.1.1 Time Speed: The Speed of game time can be controlled by player. Player can choose to pause, 1x or 2x the time flow.

[Implemented]

1.2 Money: The main currency in the game. Player earns money from operating stores, completing requests, etc. Player uses money to construct.

1.2.1 Bankrupt: At the start of Daily Settlement, if money < 0, the player will lose and need to restart from the latest checkpoint, usually the day before.

[Implemented]

1.3 Fame: A consumable resource, a second currency in game. Earned by completing requests (significantly), and in daily settlement (subtly, at the cashier), and in daily achievement. Used to upgrade product category level, buy blueprint, hold promotion event, etc. Harder to gain in the later stages.

2 Player Goal

[Not Implemented]

2.1 Store Star Upgrade: The store can be upgraded from 0 stars to 5 stars.

2.1.1 Requirement: Each upgrade requires certain requirements, and the player can upgrade in an info panel when the requirements are all met. The requirements are usually quotas to meet, not resources to be cost. For example, total shelves built, total customer amount, total shelf level, total customer level, highest historical daily income, etc.

2.1.2 Reward: Each success upgrade rewards the player with a large amount of money and fame, and a special and significant shelf or facility.

2.2 Customer Request: Customer will come to the store to make a request, and the player can click to collect them. For new customers with level 0, the request is an easy dialogue choice

and the rewards is always money, fame, or customer EXP. For regular customers, the request is always to build or upgrade a specific shelf. The rewards will have new blueprints. A request can have multiple rewards. [Archived]

[Not Implemented]

2.3 **Main Mission** (VIP Story Mission)

VIP Story Mission is the main storyline and main missions in the game. It's a customer request held by VIP customers, a kind of expanded customer service. The requirements of VIP Missions are not explicit, and players need to solve it as a puzzle through narrative context. There will only exist one active VIP Story Mission at the same time.

2.3.1 **Story:** Each VIP Story Mission has an introductory story that serves as the context of the mission's requirements. The story is shown in the form of dialogue between VIP customer and player. (See 7.7)

[Implemented]

2.4 **Daily Settlement:**

Daily settlement count and visualize the data changed during the day, including money and fame earned, served customers, completed and uncompleted achievements etc. (See 3.4.7.2)

[Not Implemented]

2.5 **Daily achievement**

- Daily achievements are tiny, random goals for the player to accomplish. There will always exist three unfinished achievements, and if any of them are achieved, new achievement requirements will be assigned randomly from the database. Status of Daily achievement is displayed during the daily settlement. (See 3.4.7.2)

3 **Store**

3.1 **Map:** Multiple unlockable areas in units of Floors. Floors are stacked and connected in an aesthetically irregular manner.

[Implemented]

3.1.1 **Floor:** Placement on the floor is grid-based. Each floor is 2-grid height. Each column can only place one shelf. Decoration and Floor connections (stairs and elevators) are placed, but don't occupy the grid.

[Not Implemented]

3.1.2 **Customer Capacity:** The limit of the number of customers that can exist in the store at a time.

[Not Implemented]

3.1.3 **Unlock New Area:** The Player can spend money to unlock new areas connected to the unlocked areas. Unlocking New Areas raises the customer capacity.

3.2 [unfinished]

3.3 **Shelves:** Placed on the floor, where customers buy products.

Shelf attributes:

3.3.1 **Price:** The customer pays the player this money per product.

3.3.2 **Maintenance Fee:** Money that's reduced at daily settlement.

3.3.3 **Stock:** The Maximum amount of products that can be bought in one day. If stock = 0, the shelf is sold out and shut down for the rest of the day. Players decide how much to restock every day. [Archived]

3.3.4 **Attractiveness:** The more attractive the shelf is, the more possible for a customer to come and buy its product. Shelf attractiveness together is the store's total attractiveness.

3.3.5 **Category:** Customer will have an interest level in each category. The customer will visit the shelf from the highest interest level to the lowest. A category can be upgraded from Lv 1 to Lv 5 with a lot of fame and money, and higher levels will need an expensive maintenance fee. Category-level upgrade significantly boosts the attractiveness of all shelves in the category. Category includes: Lingere, Condom, Vibrator, Fleshlights, Lubricant

3.3.6 **Build Fee:** A one-time money cost when placing the building.

[Not Implemented]

3.3.7 **Embarrassment Value (EV):** Add the embarrassment value to the surrounding grid.
Formula for each grid:

- Positive EV: result += min(0, EV - Distance)
- Negative EV: result += max(0, EV - Distance)

3.4 **Facilities:** Boost floor attributes or have special use. (A short list with varied effects.)

3.4.1 Facilities attributes:

- **Effect**
- **Build Fee**
- **Maintenance Fee**

3.4.2 Facilities examples:

- **Cashier:** Each floor needs at least one cashier. Starting from the second unlocked floor, each cashier needs at least one employee.
- **Air Conditioner:** Reduce embarrassment Level for all customers on the same floor.
- **ATM:** Customer with a high interest, low embarrassment, and low money bag will use the ATM to restore their money bag.
- (...)

3.4.4 **Environmental Embarrassment Value (EEB):** Each grid has an EEB, affected by the surrounding environment (shelves, facilities, decorations).

- **Store Embarrassment Heatmap:** A heat map that shows the embarrassment value of each grid.

3.4.5 **Blueprint:** Shelves and Facilities need blueprints to be built. Blueprint is obtained from completing requests or purchased in the blueprint shop with fame.

3.4.6 **Construction Mode:** Player can place, destroy, and move shelves and facilities. Item placement is grid-based.

1. **Place:** Costs a build fee and consumes one blueprint. Construction time: 15 seconds.
2. **Move:** Cost movement fee (small amount). Construction time: 5 seconds
3. **Destroy:** Return one blueprint. No money cost or returned. Instant construction.

3.4.7 **Day Loop:** A Game time day costs 30 seconds in real time. A day counter UI in the corner counts the date in the format: Day xx.

1. **Store Hours:** 12:00 to 23:00. Time flows at normal speed. Customers AI will move and buy products in the store during store hours. The player can activate any UI. The store hours can be upgraded at a mid-stage of gameplay.

2. **Daily Settlement:** UI panel pop-up with information about the last store hours, at 10 pm. During daily settlement, the time flows quickly to 11 am, the next day, then pauses

- Daily Total Sale (all sales)
- Daily Total Expenses (all maintenance fees)
- Daily Income (sales - expenses)
- Daily Total Customer
- **Fame Earned:** A reward to the player, calculated by the daily total income and the daily total customers.
- **Daily achievement** (See 2.5)
- Competition / News / ...

3.4.8 **Trend:** Every day has a different Trend, affecting the attractiveness of products.

1. **Trend Calendar:** Trend calendar records each trend in the game. It's shown to the player at the start of the day, and the player can always check it during gameplay.

3.4.9 **Store info panel:** A UI panel that keeps track of all store information.

- **Store level:** A store level challenge from 0 stars to 5 stars. Player needs to fulfill specific requirements to upgrade the store level. Upgrading the store level will unlock blueprint and customer, reward money and fame, etc.
- **Total store count**
- **Total attractiveness count**
- **Total income count**

3.4.10 **Stock:** The Maximum amount of products that can be bought in one day.

1. **Restocking:** Performed manually by the player (or automatically / semi-automatically during daily settlement). Player controls the restock of each product by a scroll bar. Previous restock amount

- 2.

4. Customers

4.1 Customer Attributes:

4.1.1 **Interest Levels (IL):** Customer has different interest levels by category. This attribute will not change.

4.1.2 **Trust Point (TP):** Trust points are slightly accumulated through successful buy, and through customer upgrades. Trust points will affect the Money bag size. Start with 0.

4.1.3 **Embarrassment Point (EP):** Each customer instance will have its own embarrassment level, separately. Start from 0 when the customer instance enters the store. It's increased or decreased by the embarrassment level on the grid they stepped on each second. When the Embarrassment level is full, the customer will stop the buying action and run away embarrassedly.

4.1.4 **Money bag (MB):** When customers use up all the money, they will leave.

4.1.5 **EXP:** When EXP is full, upgrade the customer and empty EXP. EXP is gained from completing requests and customers making purchases.

4.2 **Customer Upgrade:** Customers are automatically upgraded to the next level when their request is completed. Upgrade grants customers some Interest Levels (differ between customers). Upgraded customers visit more frequently and buy more products.

4.2.1 Customer Levels:

Customer Type	EXP cost	Buy	Total IL Reward	
New Customer	50	1 - 2	1 - 2	
Regular	100	1 - 5	2 - 3	
Member	200	1 - 7	2 - 4	
Advocate	400	1 - 10	3 - 5	

[unfinished] 目前没有限制visit, 同时只能有一个同样customer; 购买量计算 IL x Trust Level]

4.3 Customer AI:

4.3.1 **Embarrassment Point:** A customer attribute, starting with 0. Each second, add the environmental embarrassment value of the customer's standing grid to their embarrassment point. The customer will leave when the Embarrassment point is full. Embarrassment Point cannot be below 0.

4.3.2 **Spawn:** Each customer can only visit once per day. Randomly pick customers to spawn until the customer capacity limit is hit.

4.3.3 **Buy:** The range of amount of products the customer will try to purchase at one shelf.

4.3.4 **Buy Amount:** The actual amount of product the customer will try to purchase at one shelf.

4.3.5 **Buy List:** A list of shelves that the customer will probably visit.

4.3.6 **Inventory:** Each customer has an inventory. The customer will hold the products in inventory in hand. If the customer bought it, remove the inventory visual and add a shopping bag visual.

4.3.7 **Time:** Time to wait for each action. Other customers trying to interact with the occupied object need to line up and wait till they're finished.

4.3.8 **Target:** The Customer will always move towards the target.

4.3.9 Customer AI Logic:

1. **Customers randomly spawn at the entrance.** A customer spawns no less than min-visit and no more than max-visit times in a day, randomly.
2. **Set the target to the most attractive shelf** for this customer.
The selection logic:
 1. Create a list of shelves which's in the category (/categories) with the highest customer interest point.
 2. Randomly select a target shelf from the list. The weight of random selection is the attractiveness of the shelf.
3. Handle exceptions: If the way to the shelf doesn't exist, remove the shelf from the list. Move to 1. (If this happens 3 times straight, the customer is annoyed and leaves.)
4. Move to the target shelf. Wait in line.
5. **Calculate the buy amount.** Randomly select between min-buy and max-buy.
6. Handle exceptions: If $\{(buy\ amount) * (shelf\ price) > money\ bag\}$, then $\{buy\ amount = money\ bag\ shelf\ price\}$. If $buy\ amount < 0$, Customer leaves.
[unfinished]
7. Shelf stock $-=$ buy amount.
8. **Make Payment:** Customers add the buy amount of the product to the inventory. Move to the closest register. Line up, make payment.
9. $Payment = (buy\ amount) * (shelf\ price)$. Customer's money bag $-=$ payment. Player money $+=$ payment.
10. Empty player inventory. Remove the bought shelf from the customer's buying list.
11. **Move to step 2**, loop.

4.3.10 Customer AI Notes:

- 1.

4.4 Customer Actions:

Walk speed = [unfinished, need tests]. Customer speed is not changeable by the player.

Action	Time cost (seconds)	Notes
Move	Distance/speed	
Wait	(line position) * (checkout/ buy time)	
Buy	$1 + (buy\ amount * 0.1)$ second	
Checkout	$1 + (buy\ amount * 0.1)$ second	

4.5 Customer Traits:

4.5.1 Base properties

Lingere	Condom	Vibrator	Fleshlights	Lubricant	BDSM	Money bag
2	2	2	2	2	2	50

4.5.2 Trait List PoplifeSpreadSheet

5. Storytelling

5.1 Story Summary

[unfinished]

5.2 Background Setting

[unfinished]

5.3 Storytelling Tools

Story in the game will be narrated mainly through VIP Story Mission. Flavor text will help with world-building in Customer Request, Shelf info panel, Customer info panel, etc.

We are using Dialogue Tree in [Node Canvas](#) to implement dialogues into gameplay.

5.3.1 Tutorial Dialogue:

[Link to Tutorial Dialogue.](#)

[unfinished]

5.3.2 VIP Mission Dialogue:

[Link to VIP Mission Dialogue.](#)

[unfinished]

6. Art [unfinished]

6.1 Artstyle/reference

6.2 Character

6.3 Store

6.4 Scene

6.5 Technical Art

...

7. UI [unfinished]

[Link to UI Design Page \[Figma\]](#)

7.1 General UI

7.2 Runtime Indicator UI:

Pop Bubble: Overhead, indicating customer action, including getting products from shelves, purchasing, thoughts, and emotions, etc.

Broadcast: A one-line message about each customer's Purchase, rolling horizontally. Example format of each message: [customer name] bought [amount] of [product], + [sales money].

Messages: A list of messages about important notices. Text will be in three colors: Gold, White, and Red.

- **Gold:** Notification of the occurrence of something exciting
-
- **White:** Notification of the occurrence of something normal
- **Red:** Warning messages of the occurrence of something negative.
 - Customer leaves because no interesting shelves available
 - ...

Unlock/ Upgrade/ important, etc.

7.3 **Statistic Panel:**

Daily Settlement Panel:

7.4 **Construction Panel:**

7.5 **Information Panel:**

 7.5.1 Customer

 7.5.2 Shelf

 7.5.3 ...

7.6 **Request:**

 7.6.1 Request List

 7.6.2 Request Detail

7.7 **VIP Mission:**

8. Audio [unfinished]

8.1 Music

8.2 SFX

8.3 Audio Implementation

...

Update Log

Oct 8

- Big changes to the **request system** and its relationship with customers;
- Big changes to **VIP customer** -> now they drop main mission and don't consume as customer.
- Clarification of **customer capacity**, source of fame, and source of blueprint.

2.2 Customer Request: Customer will come to the store to make a request, and the player can click to collect them. For new customers with level 0, the request is an easy dialogue choice and the rewards is always money, fame, or customer EXP. For regular customers, the request is always to build or upgrade a specific shelf. The rewards will have new blueprints. A request can have multiple rewards.

2.3 Main Mission (VIP Story Mission)

VIP Story Mission is the main storyline and main missions in the game. It's a customer request held by VIP customers, a kind of expanded customer service. The requirements of VIP Missions are not explicit, and players need to solve it as a puzzle through narrative context. There will only exist one active VIP Story Mission at the same time.

2.3.1 Story: Each VIP Story Mission has an introductory story that serves as the context of the mission's requirements. The story is shown in the form of dialogue between VIP customer and player. (See 7.7)

2.4 Daily Settlement:

Daily settlement count and visualize the data changed during the day, including money and fame earned, served customers, completed and uncompleted achievements etc. (See 3.4.7.2)

2.5 Daily achievement

- Daily achievements are tiny, random goals for the player to accomplish. There will always exist three unfinished achievements, and if any of them are achieved, new achievement requirements will be assigned randomly from the database. Status of Daily achievement is displayed during the daily settlement. (See 3.4.7.2)

3.1.1 Floor: Each floor needs a cashier facility. Placement on the floor is grid-based. Each floor is 2-grid height. Each column can only place one shelf. Decoration and Floor connections (stairs and elevators) are placed, but don't occupy the grid.

3.1.2 Customer Capacity: The limit of the number of customers that can exist in the store at a time.

3.1.3 Unlock New Area: The Player can spend money to unlock new areas connected to the unlocked areas. Unlocking New Areas raises the customer capacity.

4.1.5 **EXP:** When EXP is full, upgrade the customer and empty EXP. EXP is gained from completing requests and customers making purchases.

4.3.2 **Spawn:** Each customer can only visit once per day. Randomly pick customers to spawn until the customer capacity limit is hit.

Anna's Meeting Notes

1/15

Shelf -> Stat

NPC -> Trait

-> Interest to tip shelf

New System:

-> player select price + import amount (maintenance)

-> Trend event every day

-> Item category: New category

-> Narrative

-> Implement story for each npc?

-> Educational info (book)

-> AI Campaign/ robot talkbot (Hymer 2000)

-> Art:

-> UI

-> NPC

-> Shelf

Linear Regression Function

```
double[] y = { /* 你的历史数据, 按时间顺序 */ };

double PredictLinear(double[] y)
{
    int n = y.Length;

    double sumX = 0, sumY = 0, sumXY = 0, sumXX = 0;

    for (int i = 0; i < n; i++)
    {
        double x = i + 1;
        sumX += x;
        sumY += y[i];
        sumXY += x * y[i];
        sumXX += x * x;
    }

    double slope = (n * sumXY - sumX * sumY) / (n * sumXX - sumX * sumX);
    double intercept = (sumY - slope * sumX) / n;

    double nextX = n + 1;
    return slope * nextX + intercept;
}
```